



Nachiketa Patil

MS by Research

Centre of Machine Intelligence & Data Science

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Examination	University	Institute	Year	CPI/%
MS by Research	IIT Bombay	IIT Bombay	2024-Present	7.54
B.E. [CSE]	University of Mumbai	Atharva College of Engg.	2019-23	8.86
Higher Secondary	Maharashtra State Board	Fr. Agnel Jr. College	2019	78.46%

THESIS, PROJECT & SEMINAR

- **Efficient Data Selection in Collaborative Machine Learning**
(R&D Project | Advisor: **Prof. Arjun Bhagoji**) (Jan'25–Present)
Current work:
 - Designing strategies for reliable data selection for multi-owner collaborative ML settings.
 - Implementing utility-based scoring mechanisms to evaluate data relevance before training.
 - Studying the effect of data heterogeneity on learning stability and generalization.**Future work:**
 - Develop active selection policies to reduce training cost while retaining accuracy.
 - Extend the framework to adversarial or noisy data contributors.
- **Joint vs. Two-Stage Optimization for Temporal Sequences**
(Research Project | Advisor: **Prof. Arjun Bhagoji, Prof. Francesco Bronzino(ENS Lyon)**) (Nov'25–Present)
 - Extended the NetSSM backbone with auxiliary heads to jointly model conditional event timing and symbolic semantics.
 - Conducted controlled experiments comparing Joint optimization with Two-Stage training strategies to evaluate timing calibration accuracy.
 - Synthesized high-fidelity replayable network traffic traces (PCAPs) statistically aligned with real-world traffic patterns (e.g., Netflix).

COURSE PROJECTS

- **Adversarial Robustness via Representation Similarity Regularization**, (DS 603: Robust Machine Learning)
(Guide: **Prof. Arjun Bhagoji**) (July'25–Nov'25)
 - Proposed a defense strategy replacing output-level constraints (e.g., TRADES) with representation-level similarity regularization.
 - Implemented geometric (CKA, HSIC) and neighborhood-based (CKNNA, MkNN) metrics to stabilize feature manifolds.
 - Demonstrated improved robustness on WideResNet-28-10 against PGD/APGD adversarial attacks.
- **Unified Multi-Task Image Denoising and Deblurring**, (GNR 650: Advanced Deep Learning for Image Analysis)
(Guide: **Prof. Biplab Banerjee**) (July'25–Nov'25)
 - Developed a multi-task restoration model using a frozen DINOv3 ViT backbone with LoRA adapters, enabling unified denoising and deblurring.
 - Identified negative transfer in specialist models and showed that task-conditioned multi-task learning improves generalization on OOD domains (e.g., faces).
 - Implemented full training pipeline (pre-training → fine-tuning → inference) and evaluated with PSNR/SSIM on SIDD, CBSD68, RealBlur, and Helen.
- **Robust 2D-DPO**, (CS 769: Optimization for Machine Learning)
(Guide: **Prof. Ganesh Ramakrishnan**) (Jan'25–May'25)
 - Implemented 2D-DPO, a 2-dimensional preference optimization technique for LLM alignment.
 - Proposed a novel robust formulation designed to resist noisy signal degradation.
 - Supported the claims using theoretical analysis and empirical results after testing with the Pythia 6.9B model.
- **Emotional Analysis for Indic Languages**, (CS 626: Natural Language Processing)
(Guide: **Prof. Pushpak Bhattacharya**) (Aug'24–Dec'24)
 - Trained and fine-tuned transformer-based models for multi-lingual emotion classification.
 - Performed fine-grained analysis (5 emotion classes) on the BHAAV dataset.
 - Benchmarked BERT, CNN and LSTM models, achieving performance comparable to modern commercial NLP systems.

- **CLIP Implementation from Scratch**, (CS 725: Foundations of Machine Learning)

(Guide: Prof. Preethi Jyothi)

(Aug'24–Dec'24)

- Implemented CLIP (ViT + Transformer) end-to-end in PyTorch.
- Trained on Flickr30k for text–image and image–text retrieval.
- Learned cross-modal representations using contrastive alignment losses.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, HTML, CSS, Bash
- **Libraries:** Pandas, Numpy, PyTorch, Sklearn, SciPy
- **Tools:** Git, $\LaTeX$, Hugging Face, CUDA, Weights & Biases, VS Code, HPC Cluster

WORK EXPERIENCE & INTERNSHIPS

- **Department System Administrator**, C-MInDS, IIT Bombay (Oct'24–Present)
 - Set up and maintained departmental mail server, LDAP, internal services portal, and multiple small services (e.g., room booking, helpdesk).
 - Configured GPU/CPU servers for research use, assisting in procurement, installation, and optimization of compute resources.
 - Implemented automation and optimized resource utilization workflows for servers and internal services.
 - Set up the departmental SLURM cluster for fair resource allocation and developed a web interface for seamless interaction with the system.
 - Managed physical infrastructure, including network racks, access points, UPS, and related offline systems.
- **Summer Intern**, Jawaharlal Nehru Port Trust (JNPT) (Summer 2021)
 - Developed scripts to automate daily data backup and transfer, analyzed the data, and prepared reports for operational insights.
 - Provided remote technical support to port employees for IT-related issues.
 - Performed basic maintenance of computers and printers.

POR & EXTRA-CURRICULARS

- **Department Technical Secretary**, C-MInDS, IIT Bombay (May'25–Present)
 - Assist the department with technical support and maintain the departmental helpdesk.
 - Manage mailing lists and internal portals to ensure smooth communication.
 - Conduct technical orientation sessions for freshers and support other student events.
- **Organizer**, Techconnect, C-MInDS, IIT Bombay (Dec'24 & Dec'25)
 - Led the planning and execution of poster presentations, ensuring smooth coordination and successful event delivery.
 - Collaborated with external speakers, professors, and industry professionals to deliver high-quality content to participants.
 - Managed logistics, including venue setup, catering, and communication with all event participants.
- **Social Media Head, NSS Volunteer**, Undergraduate College (2019–2022)
 - Led live streaming and managed social media content for NSS events, ensuring real-time engagement with a broad audience.
 - Coordinated content creation and engagement on platforms like Instagram and Facebook.